QUASI-COMPACT F GIVES QUASI-COMPACT FORMAL COMPLETION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof identifies the reduction of the formal completion morphism with the map between reduced induced subspaces, observes it factors as a thickening after a quasi-compact base change, and concludes with the thickenings lemma

Editorial note. The label lemma-quasi-compact-gives-quasi-compact names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- quasi-compact morphisms of formal algebraic spaces and their reduced models (Formal Spaces, Section formal-spaces-section-quasi-compact) with the identification of the reduction of a completion with the reduced induced subspace on T (Formal Spaces, Lemma formal-spaces-lemma-reduction-completion)
- quasi-compactness stable under base change (Morphisms of Spaces, Lemma spaces-morphisms-lemma-base-change-quasi-compact)
- thickening property morphisms: a thickening composed with a quasi-compact morphism stays quasi-compact (More on Morphisms of Spaces, Lemma spaces-more-morphisms-lemma-thicken-property-morphisms)

Source reference. Original lemma 0GB9: sources/stacks-chow/tags.txt, source lines 20364–20364. Prerequisite pointer invoked at proof line(s) 6077; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0AJ8: sources/stacks-chow/tags.txt, source lines 13291–13291. Prerequisite pointer invoked at proof line(s) 6074; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09ZY: sources/stacks-chow/tags.txt, source lines 12616–12616. Prerequisite pointer invoked at proof line(s) 6087; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03HF: sources/stacks-chow/tags.txt, source lines 4565–4565. Prerequisite pointer invoked at proof line(s) 6084; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In the situation above. If f is quasi-compact, then $f_{/T}$ is quasi-compact.*

Proof. (Quasi-compact morphisms of formal algebraic spaces are discussed in Formal Spaces, Section 0AJ8.) We have to show that $(X'_{/T'})_{red} \to (X_{/T})_{red}$ is a quasi-compact morphism of algebraic spaces. By Formal Spaces, Lemma 0GB9 this is the morphism $Z' \to Z$ where $Z' \subset X'$, resp. $Z \subset X$ is the reduced induced algebraic space structure on T', resp. T. It follows that $Z' \to f^{-1}Z = Z \times_X X'$ is a thickening (a closed immersion defining an isomorphism on underlying topological spaces). Since $Z \times_X X' \to Z$ is quasi-compact as a base change of f (Morphisms of Spaces, Lemma 03HF) we conclude that $Z' \to Z$ is too by More on Morphisms of Spaces, Lemma 09ZY. □